

Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A student sets up an experiment with four different food samples in test tubes labelled W, X, Y, and Z.

- To test tube W, she adds a few drops of iodine solution and observes a blue-black color.
- To test tube X, she adds copper sulphate and caustic soda, observing a violet coloration.
- Food sample Y leaves a greasy, translucent patch when crushed on a filter paper.
- Food sample Z does not show any color change with iodine or the copper sulphate mixture, but it is known to prevent scurvy. Based on these chemical observations, which of the following meals would be best represented by the combination of W, X, and Y?

- A) A bowl of boiled spinach, a glass of fresh orange juice, and an apple.
 B) A plate of boiled rice, a bowl of thick dal (lentils), and a spoonful of ghee.
 C) A piece of grilled fish, a boiled egg, and a glass of skimmed milk.
 D) A fresh cucumber salad, a slice of watermelon, and roasted peanuts.

2. Study the following statements about solids, liquids, and gases.

- P. Solids have a definite shape and volume.
 Q. Liquids have no space between their molecules.
 R. Gases can move freely and spread easily.
 S. Liquids have a fixed shape like solids.

Which of the following options correctly identifies the true (T) and false (F) statements?

	P	Q	R	S
A)	T	F	T	F
B)	T	T	F	F
C)	F	F	T	T
D)	T	F	F	T

3. Three identical cups contain water at same temperature. Hand A was in hot water, Hand B in cold water. Both are dipped into the cups. Which is correct?

- A) Both feel same temperature
 B) A feels cold, B feels hot
 C) A feels hot, B feels cold
 D) No sensation

4. In a dense tropical rainforest, tall trees form a thick canopy blocking sunlight from reaching the ground. Down on the forest floor, a specific type of creeping plant has massive, extra-broad, dark-green leaves. What is the fundamental, logical reason for this specific leaf

adaptation in the context of the plant's living characteristics?

- A) To maximize the surface area and chlorophyll concentration to capture the very faint, scattered sunlight available for photosynthesis on the dark forest floor.
 B) To capture as much heavy rainfall as possible to weigh the plant down against strong winds.
 C) To act as a protective umbrella for small insects that help pollinate the plant's underground roots.
 D) To increase the rate of transpiration to cool down the hot forest floor environment.

5. Diversity in living organisms allows us to group them into herbs, shrubs, and trees based on their stem structure and height.

- Specimen M has a green, tender, non-woody stem and is 1 meter tall.
- Specimen N has a hard, thick, woody main trunk with branches arising high up from the ground.
- Specimen O has a hard, woody stem, but branches arise directly from the base near the soil, creating a bushy appearance.

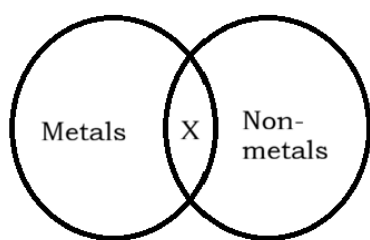
Which option correctly identifies their categories and matches them with a valid example?

- A) M = Shrub (Rose), N = Tree (Mango), O = Herb (Tomato).
 B) M = Herb (Tomato), N = Tree (Mango), O = Shrub (Lemon).
 C) M = Tree (Coconut), N = Herb (Mint), O = Shrub (Hibiscus).
 D) M = Herb (Wheat), N = Shrub (Jasmine), O = Tree (Banyan).

6. A compass is placed near a current-carrying straight wire and a bar magnet simultaneously. When the current direction is reversed, the compass needle changes direction, but not exactly opposite. Why?

A) The magnetic fields combine vectorially in different directions
 B) The bar magnet stops producing magnetic field
 C) The compass needle loses magnetism temporarily
 D) The Earth's field becomes negligible

7. Which of the following properties should be placed in the X region of Venn diagram?



A) Can be solid at room temperature
 B) Good conductors of electricity
 C) Brittle in nature
 D) Mostly gases

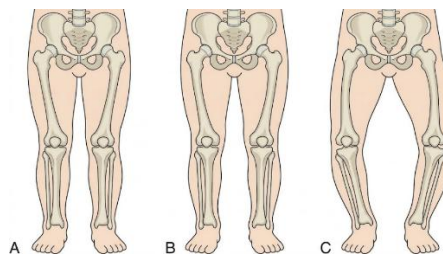
8. Which does the below diagram represent?



A) Air consists of water vapour
 B) Air consists of oxygen

C) Air consists of nitrogen
 D) Air consists of carbon dioxide

9. Observe the diagram of three different human bones. Bone A is a normal healthy leg bone. Bone B is slightly curved. Bone C is severely bent and "bow-shaped". This condition is often observed in children who lack exposure to sunlight and have a specific nutrient deficiency. Identify the nutrient and the disease.



A) Vitamin C; Scurvy. B) Iodine; Goitre.
 C) Vitamin D; Rickets. D) Iron; Anaemia.

10. A candle of height 5 cm is placed 20 cm in front of a pinhole camera. The screen is placed 5 cm behind the pinhole. The image formed is then observed in a plane mirror placed 15 cm behind the screen. Find the distance between the top of the candle and the top of the final image seen in the mirror.

A) 40.35 cm B) 55.35 cm
 C) 60.35 cm D) 70.35 cm

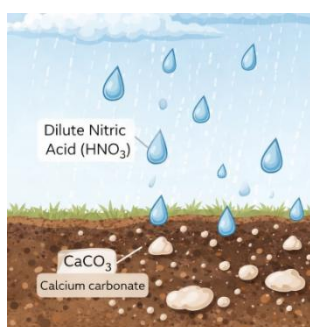
11. Observe the breathing mechanisms of the following: (1) An earthworm, (2) A fish, and (3) A human. Which statement correctly identifies how they exchange gases with their environment?

A) Earthworm: Lungs; Fish: Gills; Human: Skin.
 B) Earthworm: Lungs; Fish: Mouth; Human: Gills.

C) Earthworm: Gills; Fish: Skin; Human: Lungs.

D) Earthworm: Skin; Fish: Gills; Human: Lungs.

12. A farmer noticed the plants are growing better after rain. What might be the reason?



- A) Formation of calcium nitrate
B) Formation of calcium carbonate
C) Release of oxygen
D) Increase in nitrogen gas in air

13. When an organism dies, all its characteristics—movement, respiration, growth, response to stimuli—stop permanently. At a microscopic level, what is the most fundamental change that occurs?

- A) The organism becomes much lighter in weight because the "life" has left it.
B) The cells stop performing chemical reactions to release energy and maintain structure.
C) The organism turns into a non-living object like a stone and will never decompose.

D) The organism's DNA immediately evaporates into the air.

14. A dietician prepares a balanced diet chart for a 10-year-old child. The chart includes a large portion of energy-yielding foods, a moderate portion of body-building foods, and adequate protective foods. However, the child refuses to drink any water or eat juicy fruits, consuming only completely dehydrated dry foods.

Which of the following critical bodily functions will fail FIRST due to this dietary choice?

- A) The ability to build new muscle tissues using proteins.
B) The absorption of nutrients in the intestine, transport of digested food through the blood, and the regulation of body temperature via sweating.
C) The storage of excess carbohydrates as fats beneath the skin.
D) The mechanical breakdown of food by the teeth.

15. A car covers half the total distance at 20 m/s, one-fourth at 40 m/s, and the remaining distance at 10 m/s. Total distance is 800 m. Find the average speed.

- A) 20.2 m/s B) 17.8 m/s
C) 16.8 m/s D) 18.8 m/s

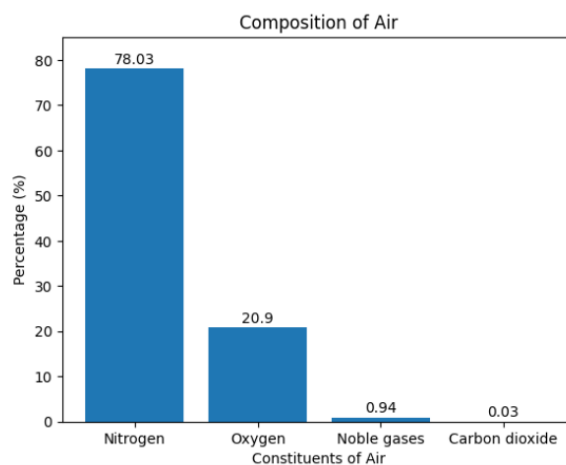
Section - 2

16. Mountain habitats present challenges like low oxygen, extreme cold, and heavy snow. Which of the following are adaptations seen in mountain organisms?

- A) Coniferous trees are cone-shaped with sloping branches to let snow slide off easily.

- B) The Snow Leopard has thick fur on its body, including its feet and toes, to protect against cold.
- C) Mountain goats have strong hooves to help them run along the rocky slopes.
- D) Trees in high altitudes have broad, thin leaves to maximize the capture of sunlight.
17. A large light source is used to form a shadow of an object. Which of the following changes will reduce the penumbra region?
- A) Reducing size of source
- B) Bringing source closer to object
- C) Increasing distance between object and screen
- D) Using a point source
18. All living organisms respond to changes in their environment. Which of the following are examples of this characteristic?
- A) A plant's stem bending toward a source of light (Phototropism).
- B) A person's pupils constricting (getting smaller) when a bright light is flashed into their eyes.
- C) The leaves of a "Touch-me-not" (*Mimosa pudica*) plant folding up when touched.
- D) A dog salivating when it hears the sound of its food bowl being filled.
19. Based on the given graph of composition of air, select the correct statements.
- A) The combined percentage of noble gases and carbon dioxide is less than 1%.
- B) Oxygen makes up approximately one-fourth of the total air composition.
- C) Nitrogen is more than three times the percentage of oxygen present in air.

D) Carbon dioxide constitutes more than



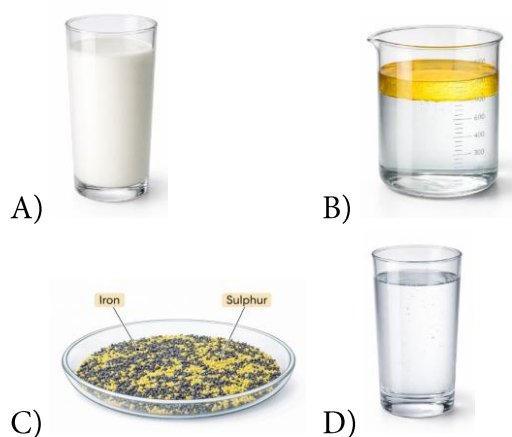
0.1% of air.

20. The camel is often called the "Ship of the Desert." Which of the following are TRUE biological facts about its adaptation?
- A) It excretes very little urine and its dung is dry to prevent water loss.
- B) The hump stores fat, which can be broken down for energy and water during long journeys.
- C) Its long legs keep its body away from the intense heat radiating from the sand.
- D) It sweats profusely throughout the day to keep its body temperature at exactly 37°C
21. An object enters Earth's atmosphere and burns, while another object near the Sun develops a tail always pointing away from the Sun. Which are correct?
- A) The burning object is a meteor
- B) The object forming a tail is a comet
- C) Comet tail direction depends on solar wind
- D) Asteroids develop long glowing tails near Earth
22. Vitamin C (Ascorbic Acid) is known to be "heat-labile." If a student prepares a salad

using freshly squeezed lemon juice but then decides to sauté the entire salad in a hot pan for 15 minutes, which of the following statements are correct?

- A) The Vitamin C content will significantly decrease due to oxidation and heat.
- B) The minerals (like Iron or Calcium) in the vegetables will also evaporate into the air.
- C) It is better to add lemon juice *after* the food has cooled down to preserve its nutritional value.
- D) Boiling or frying causes water-soluble vitamins like Vitamin C to be destroyed.

23. Which of the below diagram represent heterogeneous mixtures?



Section - 3

Comprehension – 1

A compass is placed on a table. A bar magnet is brought from the east side. Initially, the compass needle points north-south. As the magnet approaches, the needle starts turning. At one position, the needle points exactly between north and east. When the magnet is brought even

closer, the needle aligns almost completely east-west.

The magnet is then slowly moved away. The needle again passes through the intermediate direction and finally returns to north-south.

24. The position where the needle points exactly between north and east indicates:

- A) Only Earth's field acts
- B) Only magnet acts
- C) Both effects are comparable
- D) No magnetic field exists

25. When the needle aligns east-west, what is MOST correct?

- A) Earth's field becomes zero
- B) Magnet's effect dominates over Earth's field
- C) Compass loses magnetism
- D) Needle becomes fixed

26. The path followed by needle direction while moving magnet is:

- A) Random
- B) Sudden jump
- C) Continuous change
- D) No change

Comprehension - 2

Ground water filters through the pores between grains of soil and rocky material, forming underground reservoirs called X. Therefore, underground water is relatively pure and free from suspended impurities. The upper level of this stored water is known as Y. To access this underground water, structures called Z are created by digging or drilling.

27. What does X refer to in the passage?

- A) Water table
- B) Aquifer
- C) Spring water
- D) River water

28. What is represented by Y in the passage?

- A) Well water B) Water table
C) River D) Pond

29. Which of the following is used to draw water from X?

- A) Table B) Well
C) Chair D) Rock

Comprehension - 3

Living organisms persist across highly contrasting environments through intricate adaptive mechanisms shaped by long-term evolutionary pressures. These adaptations may manifest structurally (morphological modifications), behaviorally (activity patterns or responses), or physiologically (internal biochemical or metabolic adjustments). For example, xerophytic plants such as cacti exhibit reduced leaf surface area and modified tissues to minimize transpiration, while certain animals adopt nocturnality to circumvent thermal stress and conserve energy. Importantly, such adaptive traits are not immediate responses but arise through the gradual selection of advantageous variations over successive generations, ensuring survival and reproductive success within specific ecological niches.

30. The emergence of adaptive traits in organisms is primarily a result of:

- A) Sudden environmental necessity
B) Gradual selection over generations
C) Individual effort during lifetime
D) Random daily changes

31. The reduction of leaf surface area in desert plants is mainly associated with:

- A) Enhanced photosynthetic rate
B) Minimization of water loss

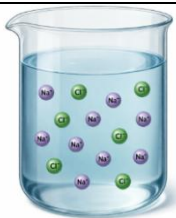
- C) Increased nutrient absorption
D) Rapid reproduction

32. Nocturnal behavior in certain animals is an adaptation that primarily:

- A) Maximizes exposure to sunlight
B) Reduces competition for food only
C) Helps avoid extreme environmental stress
D) Enhances immediate growth rate

Section - 4

33. Match Column A with Column B

Separation Technique	Used For
(A) Decantation	(P) 
(B) Sublimation	(Q) 
(C) Handpicking	(S) 
(D) Evaporation	(R) 

- A) A-P, B-Q, C-R, D-S
B) A-Q, B-P, C-R, D-S
C) A-R, B-S, C-P, D-Q
D) A-Q, B-S, C-R, D-P

34. Match Column A with Column B

Column A	Column B
(A) Carbohydrates	(P) Protective food (protects against diseases)
(B) Proteins	(Q) Energy-giving food (main source)
(C) Vitamins & Minerals	(R) Concentrated energy source (stored energy)
(D) Fats	(S) Body-building food (growth and repair)

- A) A–Q, B–S, C–P, D–R
 B) A–R, B–Q, C–S, D–P
 C) A–Q, B–P, C–S, D–R
 D) A–S, B–Q, C–P, D–R

35. Match Column A with Column B

Column A (Situation)	Column B (Correct Combined Result)
(A) Body completes one full circular revolution	(P) Distance = circumference, displacement = 0
(B) Body completes half circular path	(Q) Distance = πr , displacement = $2r$
(C) Body moves equal distance in equal time	(R) Uniform speed, average speed = constant
(D) Body moves but direction keeps changing	(S) Velocity changes even if speed constant

- A) A–P, B–Q, C–R, D–S
 B) A–Q, B–P, C–S, D–R
 C) A–P, B–R, C–Q, D–S
 D) A–S, B–Q, C–R, D–P